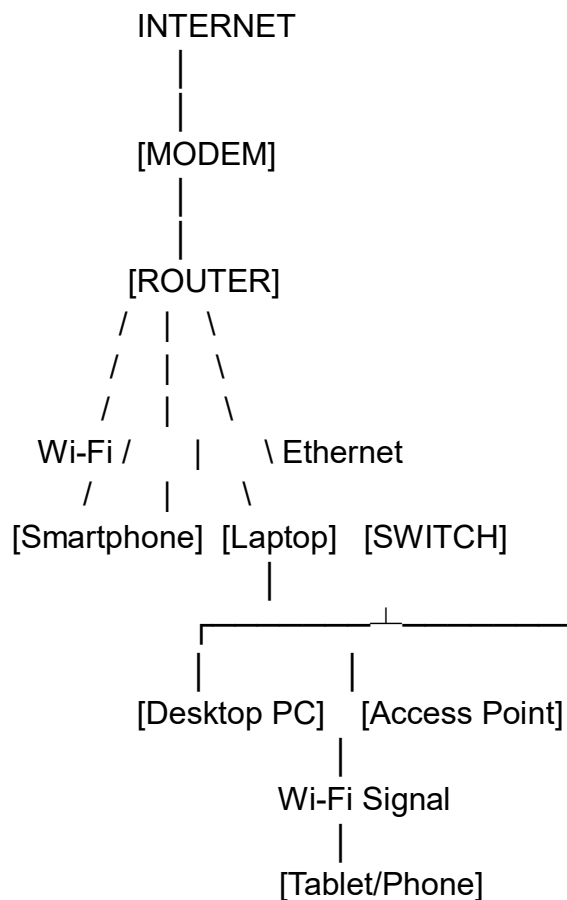


Introduction to Networking Hardware

Networking hardware devices work together to connect computers, smartphones, and other devices so they can communicate and access the internet. In a typical home or office network, the modem connects to the Internet Service Provider (ISP), the router manages traffic between devices and the internet, switches expand the number of wired connections, access points provide wireless connectivity, and network cables carry data between devices.

Task 1: Labeled Diagram of a Typical Home Wi-Fi Setup



How it works:

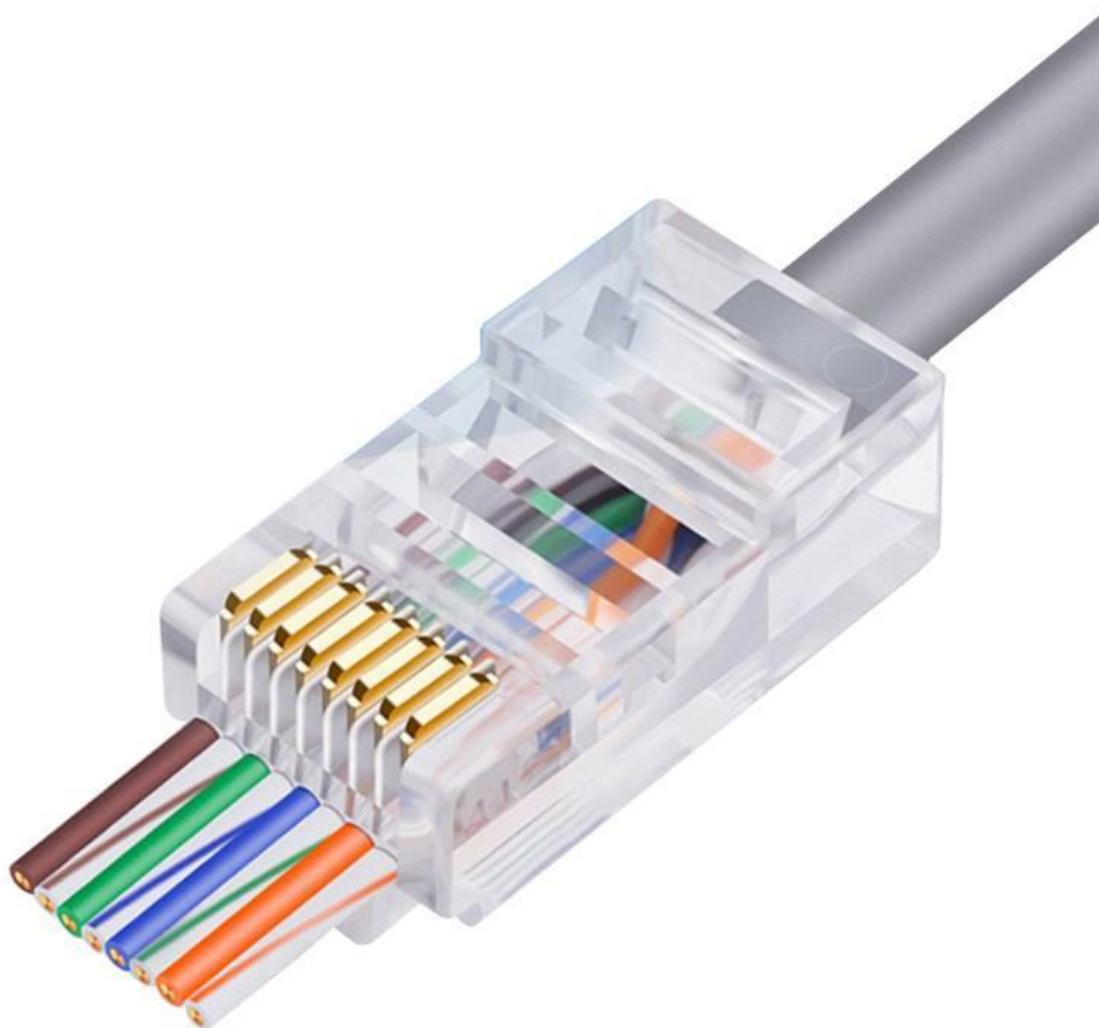
- **Modem** connects the home to the ISP.
 - **Router** directs traffic between the internet and local devices.
 - **Switch** adds more wired network ports.
 - **Access Point (AP)** extends wireless coverage.
 - **Laptop and Smartphone** connect to the network.
-

Task 2: Functions of Networking Hardware

Device	Main Function	Real-Life Example
Modem	Connects a home or business network to the ISP and converts signals for internet access.	Home broadband connection from a cable or fiber provider.
Router	Routes data between devices and the internet; often provides Wi-Fi.	Home Wi-Fi router used to share internet with family devices.
Switch	Connects multiple wired devices within a local network.	Office network connecting many desktop computers.
Access Point (AP)	Provides wireless network access and extends Wi-Fi coverage.	School or hotel Wi-Fi system.
Network Cable	Physically carries data between network devices.	Ethernet cable connecting a PC to a router.

Task 3: Network Cable Types with Images and Sources

1. Ethernet Cable (RJ45)



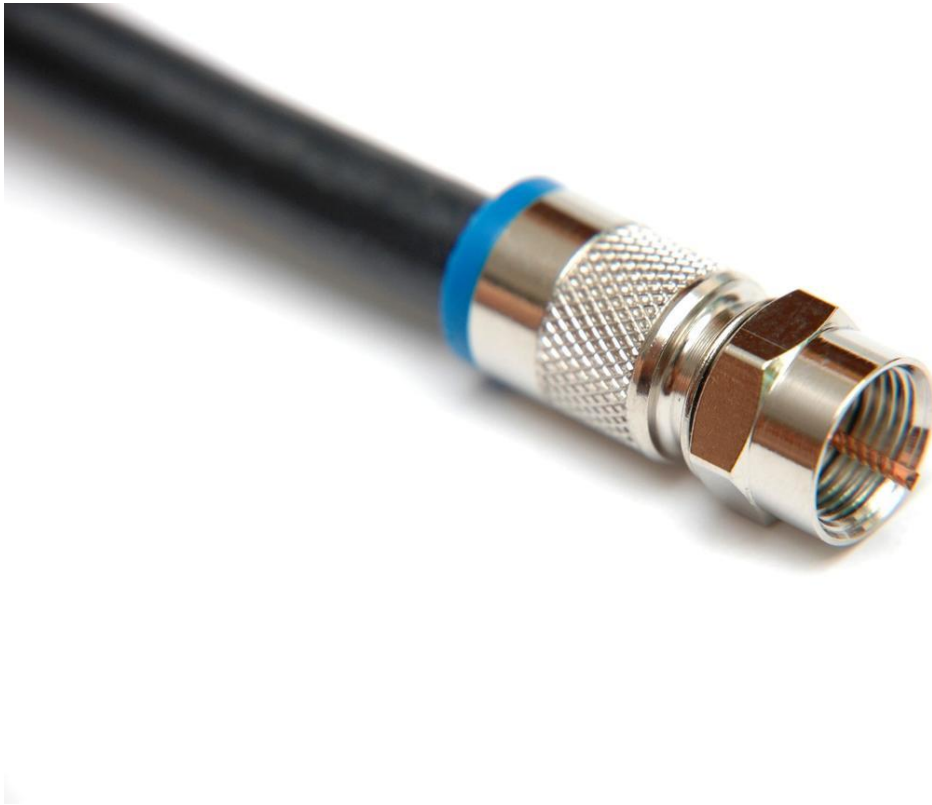




Common Use: Used to connect computers, routers, switches, printers, and gaming consoles in a wired network.

Source:

2. Coaxial Cable







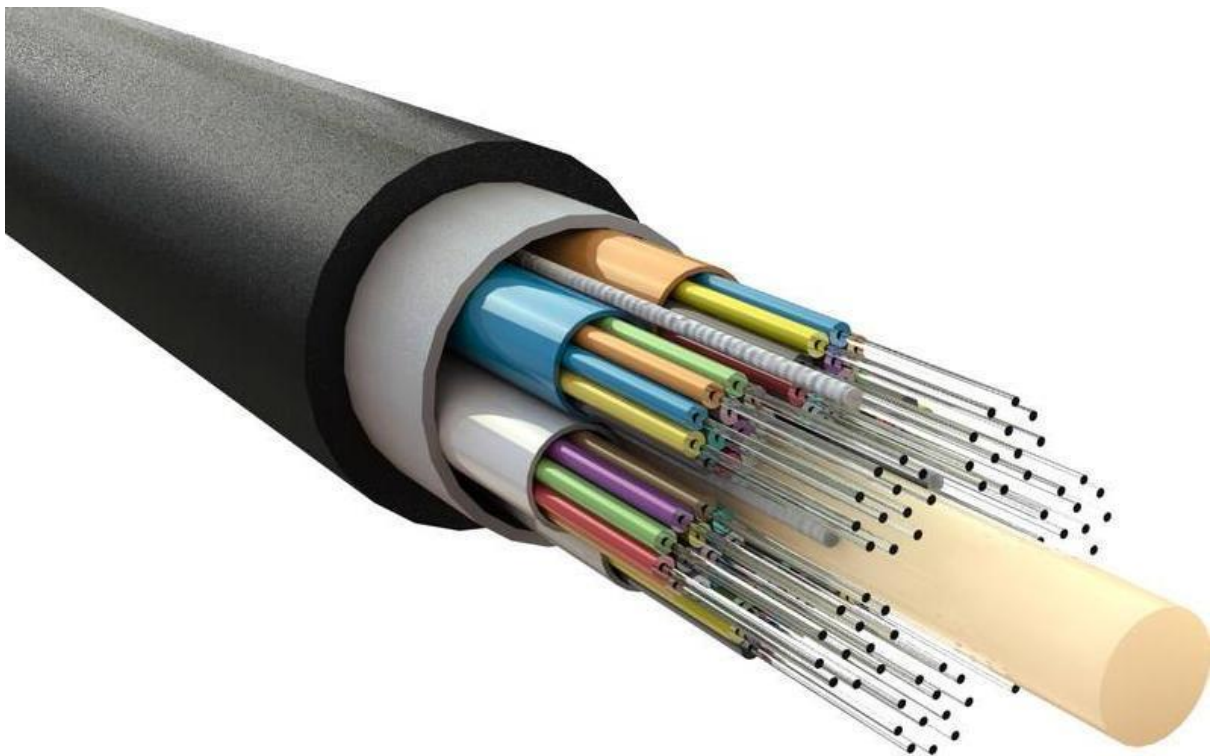
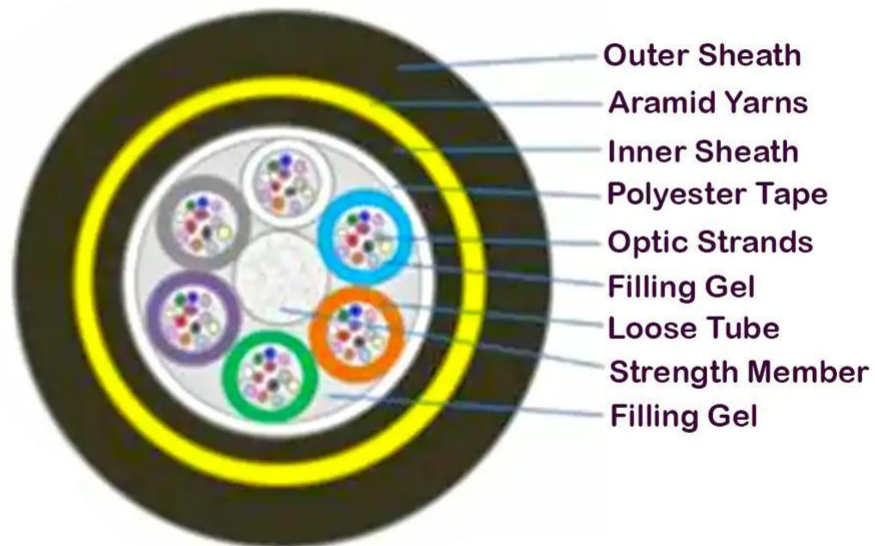
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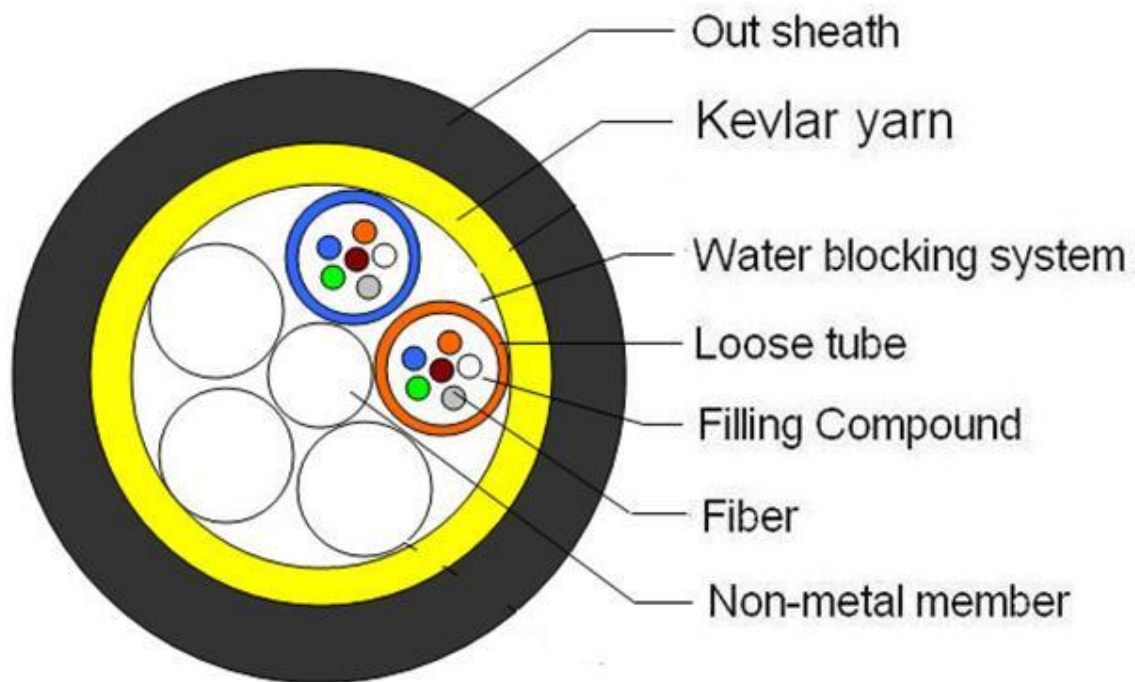
Common Use: Used by cable TV providers and some cable internet connections to deliver signals into homes.

Source:

3. Fiber Optic Cable

Cross Section Of Fiber Optic Cable





5

Common Use: Used by high-speed internet providers and telecommunications networks to transmit data using light signals.

Source:

Task 4: Router vs Switch Comparison Table

Feature	Router	Switch
Main Purpose	Connects different networks (LAN to Internet).	Connects devices within the same LAN.

Feature	Router	Switch
Typical Location	Home gateway or office network edge.	Inside homes, schools, and offices where extra wired ports are needed.
Data Handling	Uses IP addresses to determine where data should go.	Uses MAC addresses to forward data directly to devices.
Internet Sharing	Shares internet access among multiple devices.	Does not provide internet access by itself.
Wireless Support	Often includes Wi-Fi functionality.	Usually wired only.
Security Features	Can provide firewall, NAT, and traffic filtering.	Generally provides little or no security features.
Network Layer	Operates mainly at Layer 3 (Network Layer).	Operates mainly at Layer 2 (Data Link Layer).

Summary

- A **modem** connects your network to the internet.
- A **router** manages traffic and shares internet access.
- A **switch** connects multiple wired devices.
- An **access point** provides Wi-Fi coverage.
- **Network cables** carry data between devices.
- Routers connect networks together, while switches connect devices within the same network.